SCALES

The proportion by which the drawing of a given object is enlarged or reduced is called the scale of the drawing.

The scale of a drawing is indicated by a ratio, called the *Representative Fraction* (RF) or *Scale Factor*. RF is a ratio of the length of an object on a drawing to the actual length of the object.

Representative Fraction (R.F.)= (Length on drawing)/(Actual length)

The terms 'scale' and 'RF' are synonymous

Commonly expressed in the format $X : Y$

FOR FULL SIZE SCALE
R.F.=1 OR (1:1)
MEANS DRAWING
& OBJECT ARE OF
SAME SIZE.
Other RFs are described
as
1:10, 1:100,
1:1000, 1:1,00,000

Enlarging or Enlargement Scales

When smaller objects are to be drawn, they often need to be enlarged

i.e., Tiny objects dimensions must be increased, Hence this scale is called ENLARGING SCALE.

Length of an object on the drawing > actual length of the object.

Enlarging scales are mentioned in the format ***X* : 1**, where $X > 1$; **RF > 1**

Applications :

Screws and gears used in small electronic gadgets, wristwatch parts, resistors, transistors, ICs.

Reducing or Reduction Scales

Dimensions of large objects must be reduced to accommodate on standard size drawing sheet. This reduction creates a scale of that reduction ratio, which is generally a fraction. Such a scale is called **REDUCING SCALE.**

Length of the object on the drawing < actual length of the object.

Reducing scales are mentioned in the format **1 :*Y***, where $Y > 1$; **RF < 1**

Applications : multi-storeyed buildings, bridges, huge machinery, ships etc.

SCALES

BE FAMILIAR WITH THESE UNITS.

1 KILOMETRE	= 10 HECTOMETRES
1 HECTOMETRE	= 10 DECAMETRES
1 DECAMETRE	= 10 METRES
1 METRE	= 10 DECIMETRES
1 DECIMETRE	= 10 CENTIMETRES
1 CENTIMETRE	= 10 MILLIMETRES

USE FOLLOWING FORMULAS FOR THE CALCULATIONS IN THIS TOPIC.

A REPRESENTATIVE FACTOR (R.F.) =

$$\frac{\text{DIMENSION OF DRAWING}}{\text{DIMENSION OF OBJECT}}$$

$$= \frac{\text{LENGTH OF DRAWING}}{\text{ACTUAL LENGTH}}$$

$$= \sqrt{\frac{\text{AREA OF DRAWING}}{\text{ACTUAL AREA}}}$$

$$= \sqrt[3]{\frac{\text{VOLUME AS PER DRWG.}}{\text{ACTUAL VOLUME}}}$$

B LENGTH OF SCALE = R.F. X MAX. LENGTH TO BE MEASURED

Scales commonly used by engineers

1. **Plain Scales**
   - Used to read lengths in two units or to read to first decimal accuracy
2. **Vernier Scales**
   - Used to measure small divisions up to three units or to read to two decimal accuracy
3. **Diagonal Scales**
   - Used to measure in three units such as dm, cm and mm

Construction of scales : General procedure

- **All the above scales are constructed by drawing a line of length equivalent to the actual distance to be represented.**
  - **This length is called *length of scale* (LOS).**
- **LOS is calculated by the formula**

LOS = RF X Maximum distance to be represented

> - **LOS is usually calculated in terms of millimetre/centimeter.**
> - **If the maximum distance to be represented is not known, it may be taken equal to the maximum measurement (rounded off to the higher whole number) to be made with the help of the scale.**

PLAIN SCALE:-This type of scale represents two units or a unit and it's sub-division

PLAIN SCALE

PROBLEM NO.1:- Draw a scale 1 cm = 1m to read decimeters, to measure maximum distance of 6 m. Show on it a distance of 4 m and 6 dm.

CONSTRUCTION:-

a) Calculate R.F.= $\frac{\text{DIMENSION OF DRAWING}}{\text{DIMENSION OF OBJECT}}$

R.F.= 1cm/ 1m = 1/100

Length of scale = R.F. X max. distance

= 1/100 X 600 cm

= 6 cm

b) Draw a line 6 cm long and divide it in 6 equal parts. Each part will represent larger division unit.

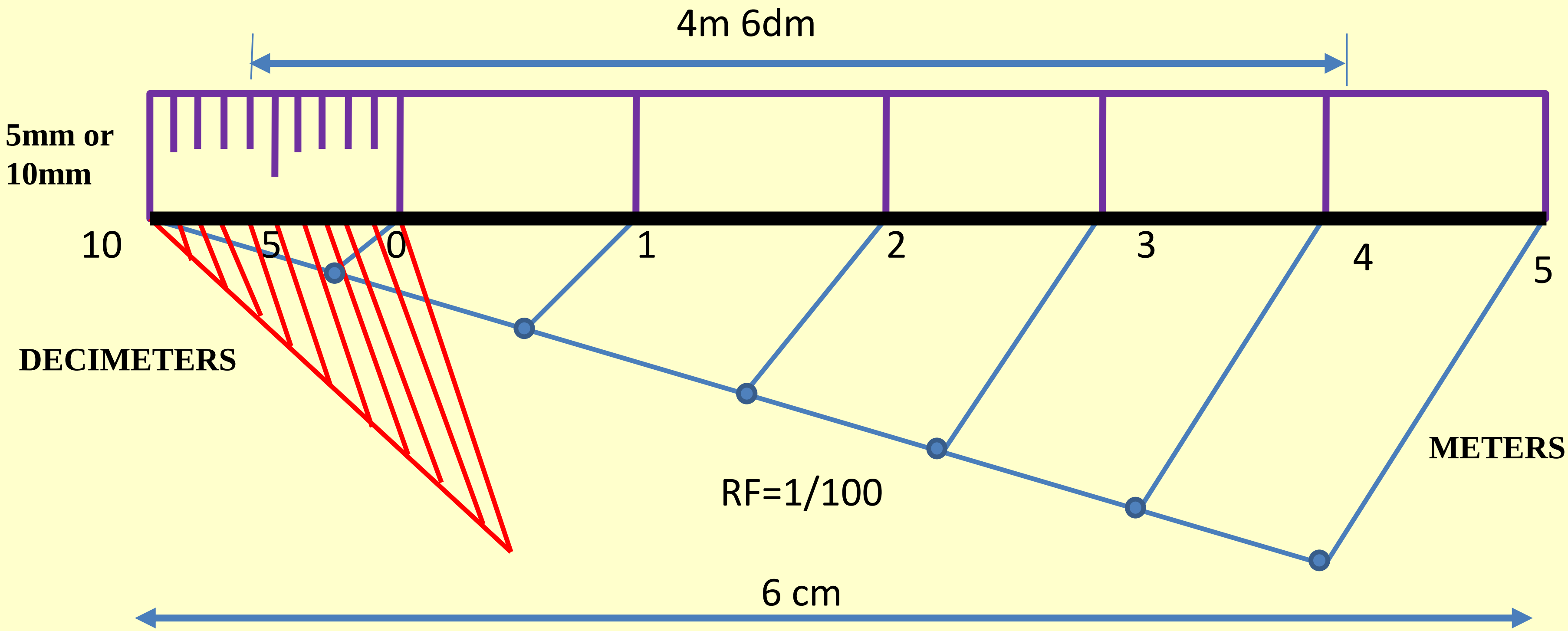
c) Sub divide the first part which will represent second unit or fraction of first unit.

d) Place (0) at the end of first unit. Number the units on right side of Zero and subdivisions on left-hand side of Zero. Take height of scale 5 to 10 mm for getting a look of scale.

e) After construction of scale mention it's RF and name of scale as shown.

f) Show the distance 4 m 6 dm on it as shown.

PLAIN SCALE:-This type of scale represents two units or a unit and it's sub-division



PLAIN SCALE SHOWING METERS AND DECIMETERS.

Q1. Draw a scale of 1cm = 1m to read decimeters, to measure maximum distance of 6m. Show on it a distance of 4m and 6 dm.

Soln:

4.6 m

4 + 0.6

4 m + 6 dm

1) $RF = \frac{1\,cm}{1\,m} = \frac{1}{100}$

2) $LOS = RF \times$ Max. distance of measurement

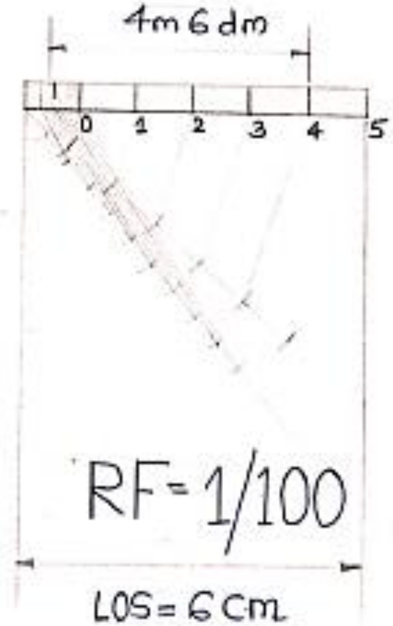
$= \frac{1}{100} \times 6\,m$

$= \frac{1}{100} \times 600\,cm$

$= 6\,cm$

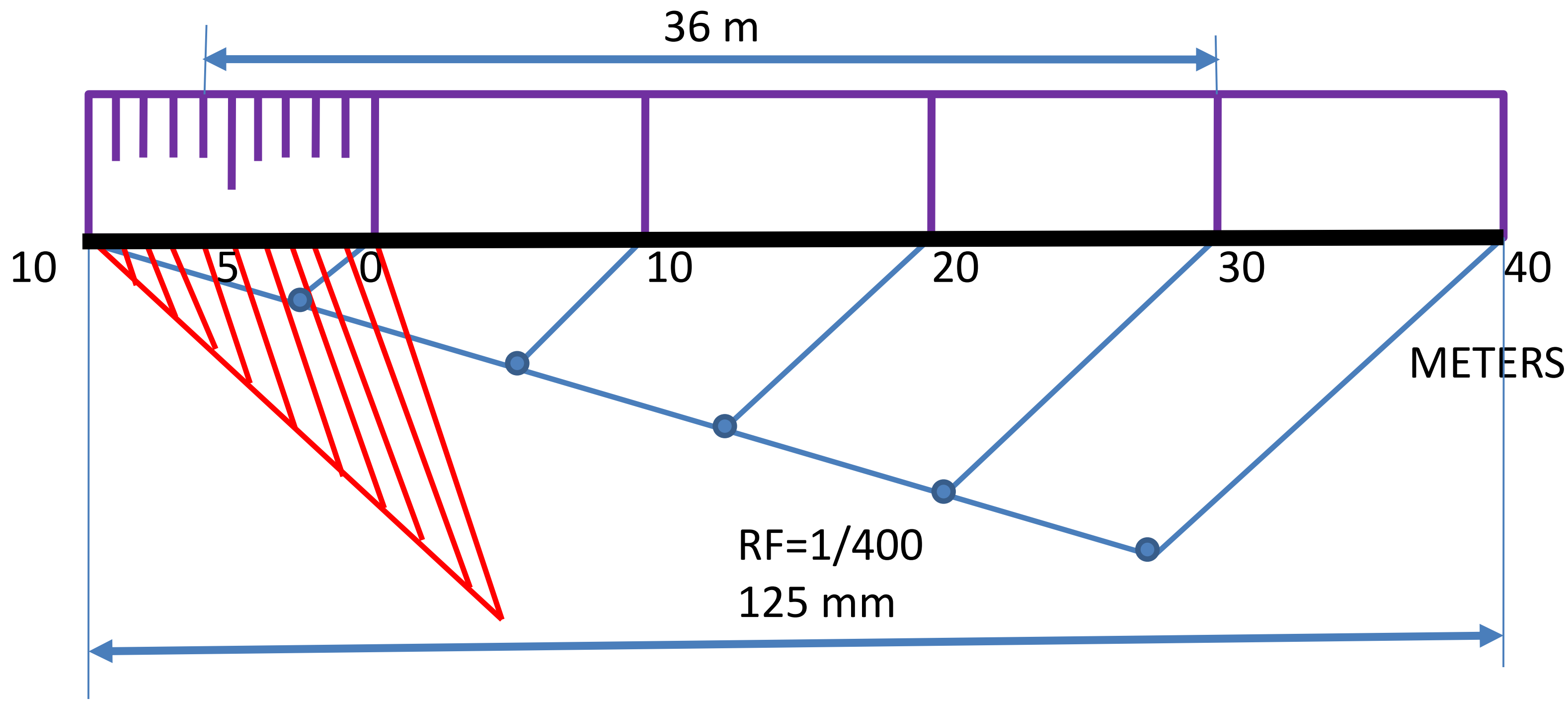
3) Divide LOS into 6 parts.

DECIMETERS



METER

PLAIN SCALE:-This type of scale represents two units or a unit and it's sub-division



PROBLEM NO.2: Construct a **plain scale** to show meters when 1 centimeter represents 4 meters and long enough to measure upto 50 meters. Find the R.F and mark on your scale a distance of 36 meters

Step 1: Calculating RF

$$R.F = \frac{Drawing\ size}{Actual\ size}(in\ same\ units) = \frac{1\,cm}{4\times 100\,cm} = \frac{1}{400}$$

Step 2: Calculating LOS

Length of the scale (LOS) = RF x Maximum distance to be measured

$$\text{LOS} = \frac{1}{400}\times 50\times 1000\ mm = 125\ mm$$

Step 3: Draw LOS as line

Step 3: Division of LOS depending upon maximum distance to be measured

PROBLEM NO.3:- In a map, a 36 km distance is shown by a line 45 cms long. Calculate the R.F. and construct a plain scale to read kilometers and hectometers, for max. 12 km. Show a distance of 8.3 km on it.

PLAIN SCALE

CONSTRUCTION:-

a) Calculate R.F.

R.F.= 45 cm/ 36 km = 45/ (36 x 1000 x 100) = 1/ 80,000

Length of scale = R.F. **X** max. distance

= 1/ 80000 **X** 12 km

= 15 cm

b) Draw a line 15 cm long and divide it in 12 equal parts. Each part will represent larger division unit.

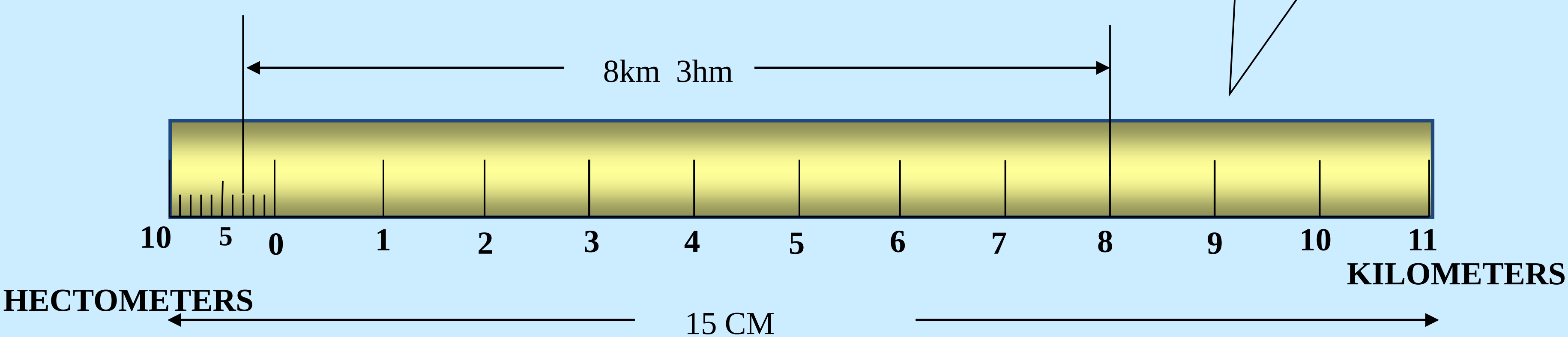
c) Sub divide the first part which will represent second unit or fraction of first unit.

d) Place (0) at the end of first unit. Number the units on right side of Zero and subdivisions on left-hand side of Zero. Take height of scale 5 to 10 mm for getting a look of scale.

e) After construction of scale mention it's RF and name of scale as shown.

f) Show the distance 8.3 km on it as shown.

for dividing the LOS, use construction technique as discussed in problem 1 and 2



R.F. = 1/80,000

PLANE SCALE SHOWING KILOMETERS AND HECTOMETERS

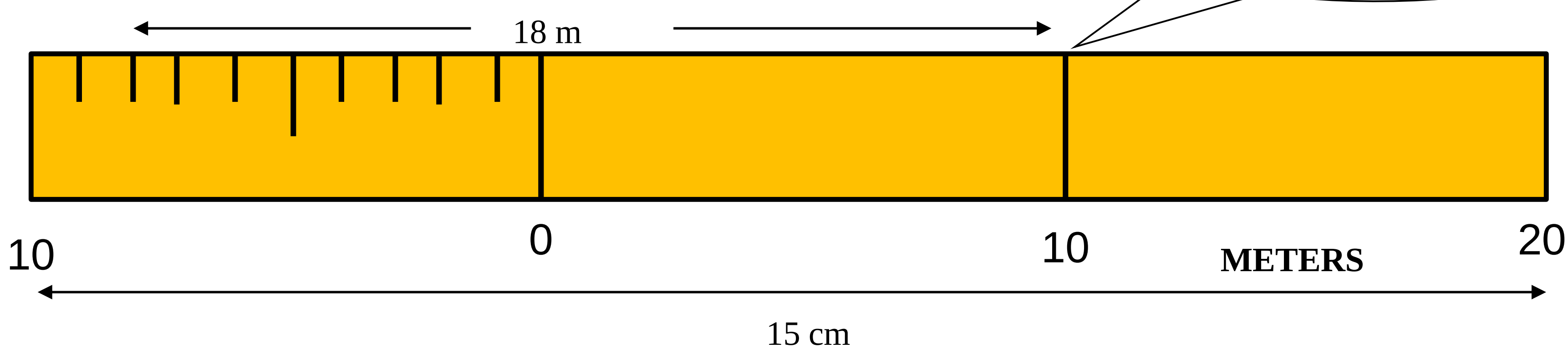
PROBLEM NO.4: A room of 1000 m^3 volume is represented by a block of 125 cm^3 volume. Find the R.F and construct a plain scale to measure up to 30 m. Measure a distance of 18m on your scale.

125 cm^3 represents 1000 m^3,

i.e., 5 cm represents 10 m, R.F. = 5 cm/10 m=1/200

LOS = RF x 30 x 100 cm = 1/200x 30x100= 15 cm

**for dividing the LOS, use construction technique as discussed in problem 1 and 2*



Different types of problems

- Construct a vernier scale of least count (LC) 2 mm. The LOS is 20 cm. Show the following distance on it 13.4 cm and 2.8 cm
- Draw a full size diagonal scale to show 0.1 mm and long enough to measure up to 5 cm. Show on this scale the following distance. (a) 0.1 mm (b) 2.35 cm and (c) 4.89 cm

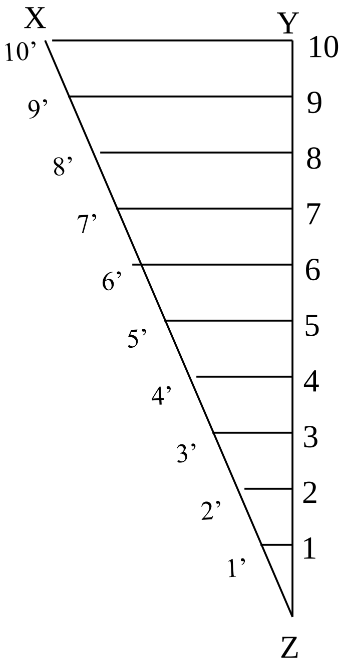
DIAGONAL SCALE

The diagonal scales give us three successive dimensions that is a unit, a sub-unit and a subdivision of a sub-unit.

Principle of Diagonal Scale: The construction of a diagonal scale is based on the principle of similarity of triangles.

- Let the XY in figure be a subunit
- From Y draw a perpendicular YZ to a suitable height
- Join XZ
- Divide YZ in to 10 equal parts
- Draw parallel lines to XY from all these divisions and number them

Consider two similar triangles XYZ and 7'7Z,
we have 7Z / YZ = 7’7 / XY (each part being one unit)
means 7'7 = 7 / 10 x XY = 0.7 XY
Similarly
1’ – 1 = 0.1 XY
2’ – 2 = 0.2 XY
Thus, it is very clear that, the sides of small triangles, which are parallel to divided lines, become progressively shorter in length by 0.1 XY.



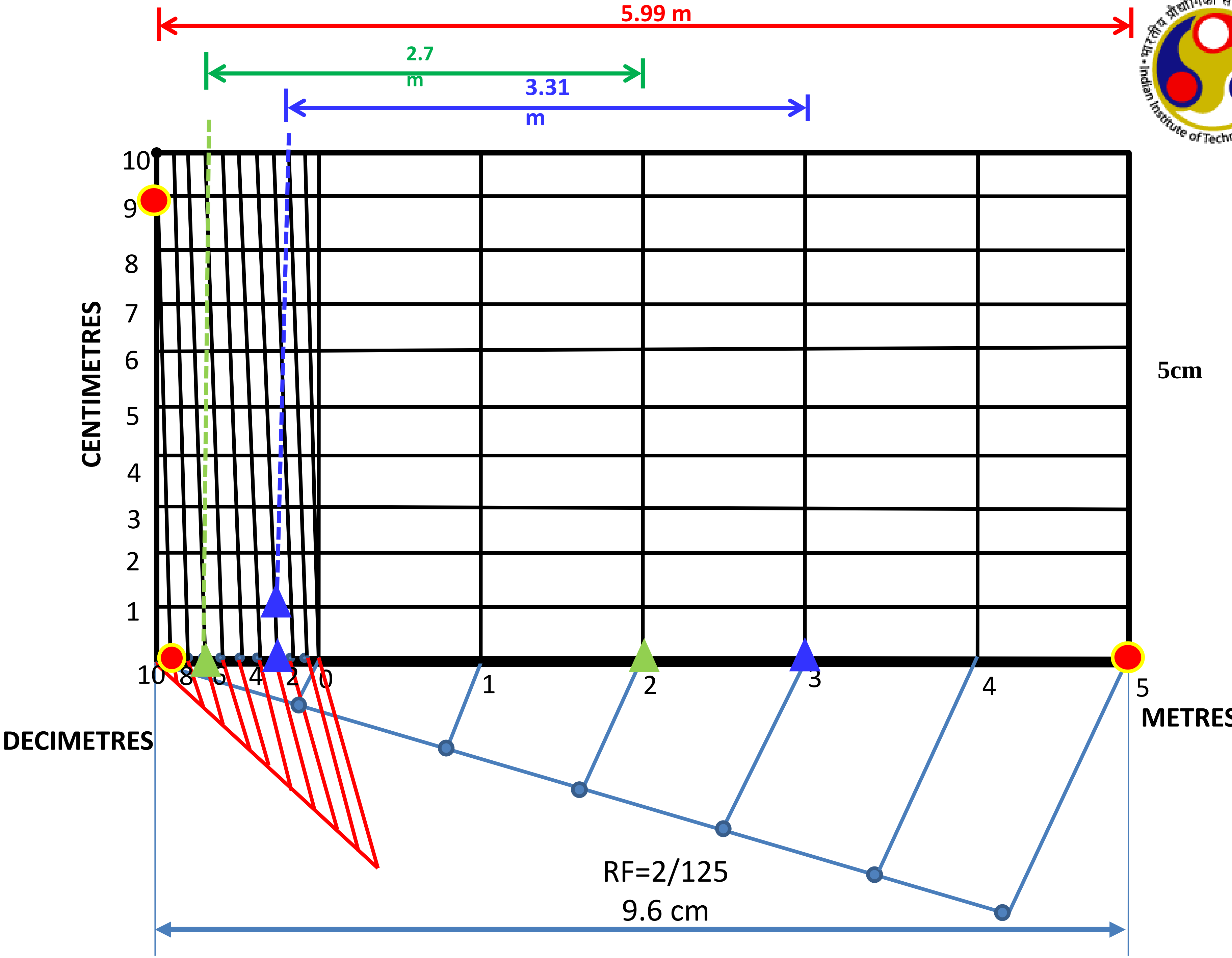
Problem 1: Construct a diagonal scale of RF= 2/125 and LC of 1 cm. Show the lengths of 5.99 m, 3.31 m and 2.7 dm on it

- Max length to be measured= 6m
- LOS = RF x 6m =2/125x600 cm= 9.6 cm

5.99= 5m+0.9 m+0.09 m=5m+9 dm+9 cm

3.31= 3m+0.3 m+0.03 m=3m+3 dm+1 cm

2.7= 2m+0.7 m=2m+7 dm



5.99 m
2.7 m
3.31 m
CENTIMETRES
10
9
8
7
6
5
4
3
2
1
5cm
10 8 6 4 2 0
1
2
3
4
5
METRES
DECIMETRES
RF=2/125
9.6 cm

Q1 Construct a diagonal scale of RF= 2/125 and LC of 1 cm. Show the lengths of 5.99m, 3.31m and 2.7dm on it. Maximum length to be measured is 6m

Soln: $RF = \frac{2}{125}$

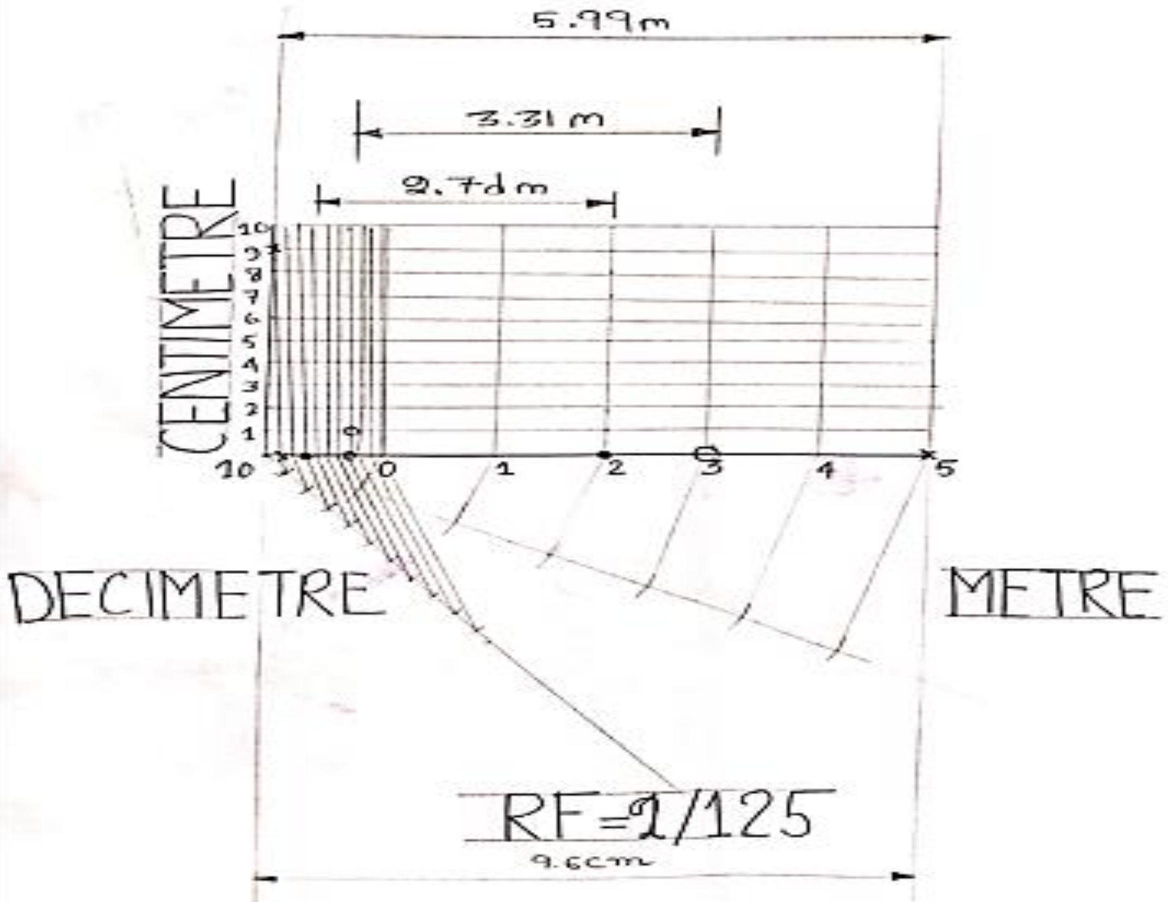
∴ Length of scale (LOS) = RF × Max length to be measured

$= 2/125 \times 6 \times 100$ cm $= 9.6$cm

5.99 m = 5m + 0.9m + 0.09m = 5m + 9dm + 9cm

3.31m = 3m + 0.3m + 0.01m = 3m + 3dm + 1cm

2.7 dm = 2 m + 0.7 m = 2m + 7dm.



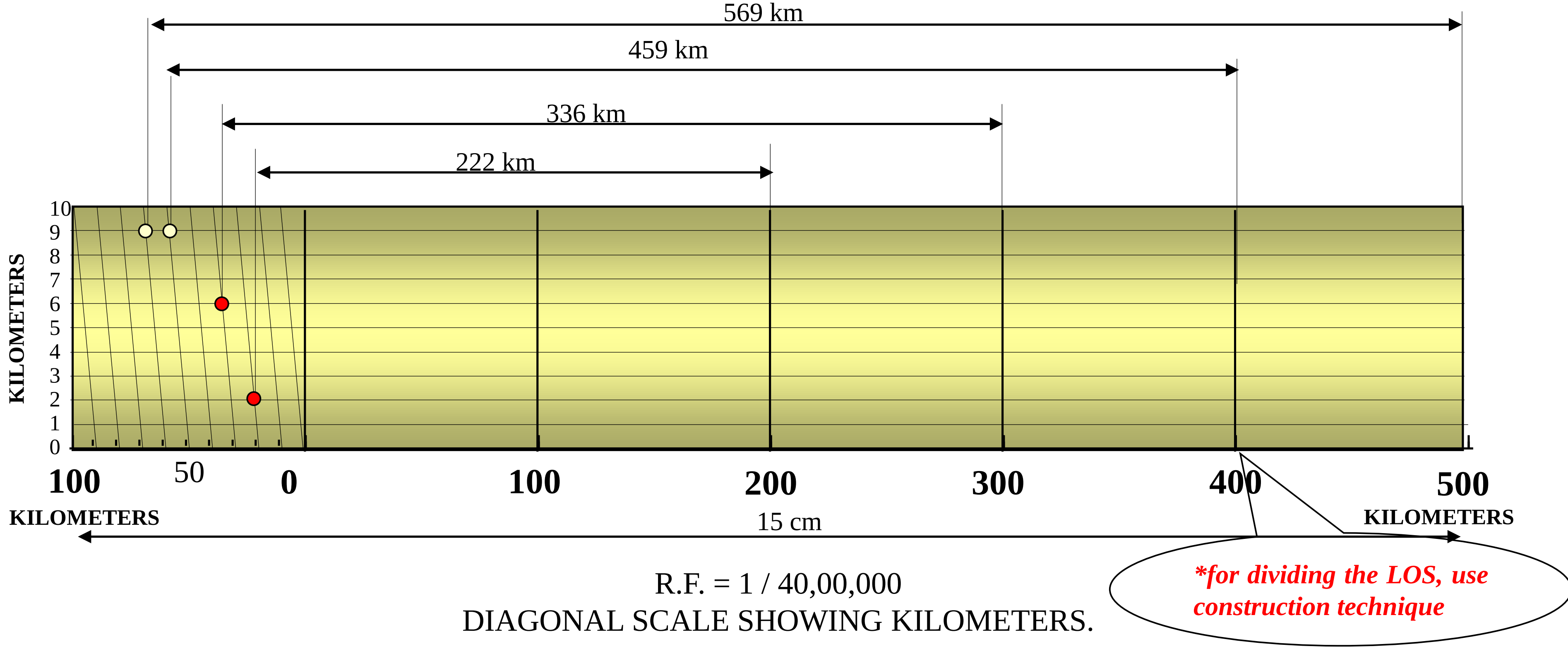
PROBLEM No. 2 : The distance between Delhi and Agra is 200 km.
In a railway map, it is represented by a line 5 cm long. Find it's R.F.
Draw a diagonal scale to show single km, and maximum 600 km.
Indicate on it following distances. 1) 222 km 2) 336 km 3) 459 km 4) 569 km

DIAGONAL SCALE

SOLUTION STEPS: RF = 5 cm / 200 km = 1 / 40,00,000

Length of scale = 1 / 40,00,000 x 600 x 10^5 = 15 cm

Draw a line 15 cm long. It will represent 600 km.Divide it in six equal parts.(each will represent 100 km.) **Divid**e first division in ten equal parts.Each will represent 10 km.**Draw** a line upward from left end and mark 10 parts on it of any distance. **Name** those parts 0 to 10 as shown.Join 9^{th} sub-division of horizontal scale with 10^{th} division of the vertical divisions. **Then** draw parallel lines to this line from remaining sub divisions and complete diagonal scale.



R.F. = 1 / 40,00,000
DIAGONAL SCALE SHOWING KILOMETERS.

PROBLEM No. 3: A rectangular plot of land measuring 1.28 hectors is represented on a map by a similar rectangle of 8 sq. cm. Calculate RF of the scale. Draw a diagonal scale to read single meter. Show a distance of 438 m on it.

DIAGONAL SCALE

SOLUTION :

1 hector = 10, 000 sq. meters

1.28 hectors = 1.28 x 10,000 sq. meters

= $1.28 \times 10^4 \times 10^4$ sq. cm

8 sq. cm area on map represents

= $1.28 \times 10^4 \times 10^4$ sq. cm on land

1 cm sq. on map represents

= $1.28 \times 10^4 \times 10^4 / 8$ sq cm on land

1 cm on map represent

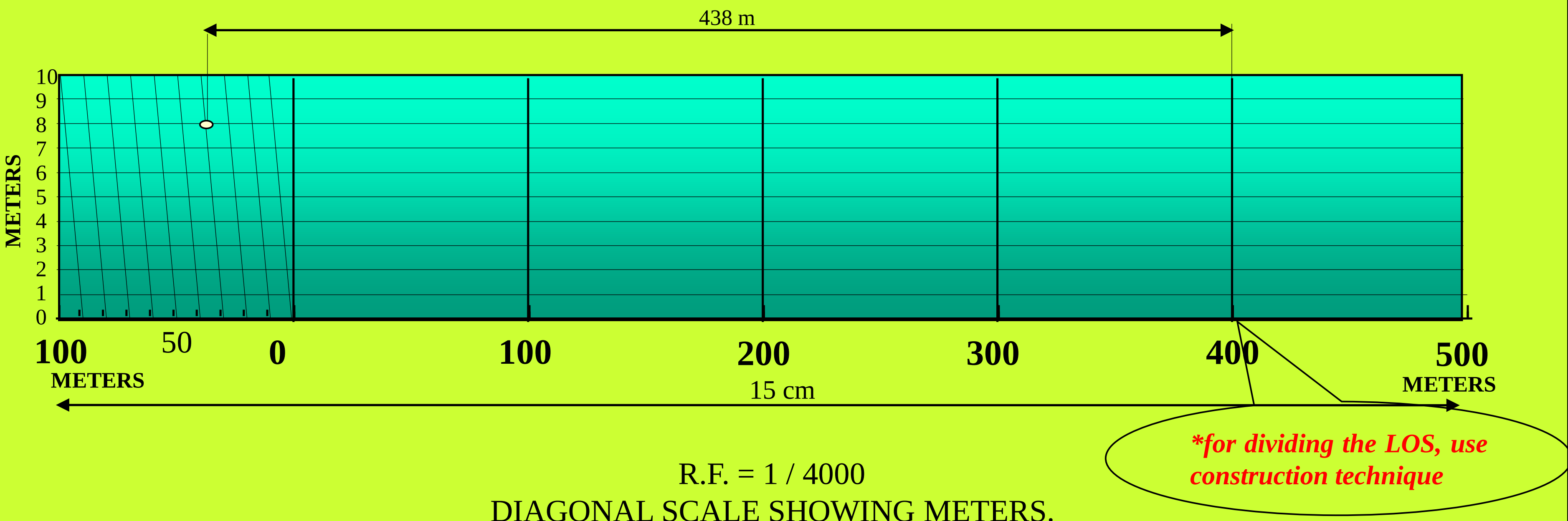
$$= \sqrt{1.28 \times 10^4 \times 10^4 / 8} \text{ cm}$$

= 4, 000 cm

1 cm on drawing represent 4,000 cm, Means RF = 1 / 4000

Assuming length of scale 15 cm, it will represent 600 m.

Draw a line 15 cm long.
It will represent 600 m.Divide it in six equal parts.
(each will represent 100 m.)
Divide first division in ten equal parts.Each will represent 10 m.
Draw a line upward from left end and mark 10 parts on it of any distance.
Name those parts 0 to 10 as shown.Join 9th sub-division of horizontal scale with 10th division of the vertical divisions.
Then draw parallel lines to this line from remaining sub divisions and complete diagonal scale.



R.F. = 1 / 4000

DIAGONAL SCALE SHOWING METERS.

PROBLEM No. 4:. Draw a diagonal scale of R.F. 1:2.5, showing centimeters and millimeters and long enough to measure up to 20 centimeters.

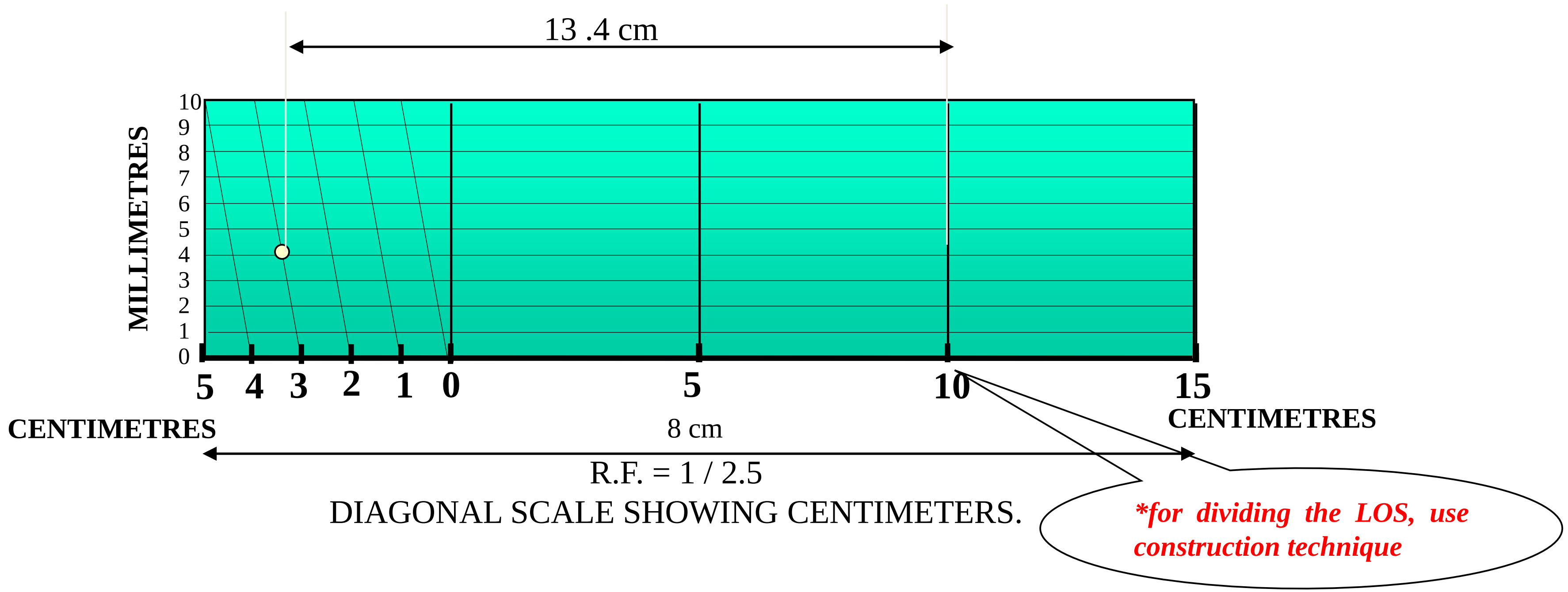
SOLUTION STEPS:

R.F. = 1 / 2.5

Length of scale = 1 / 2.5 X 20 cm.

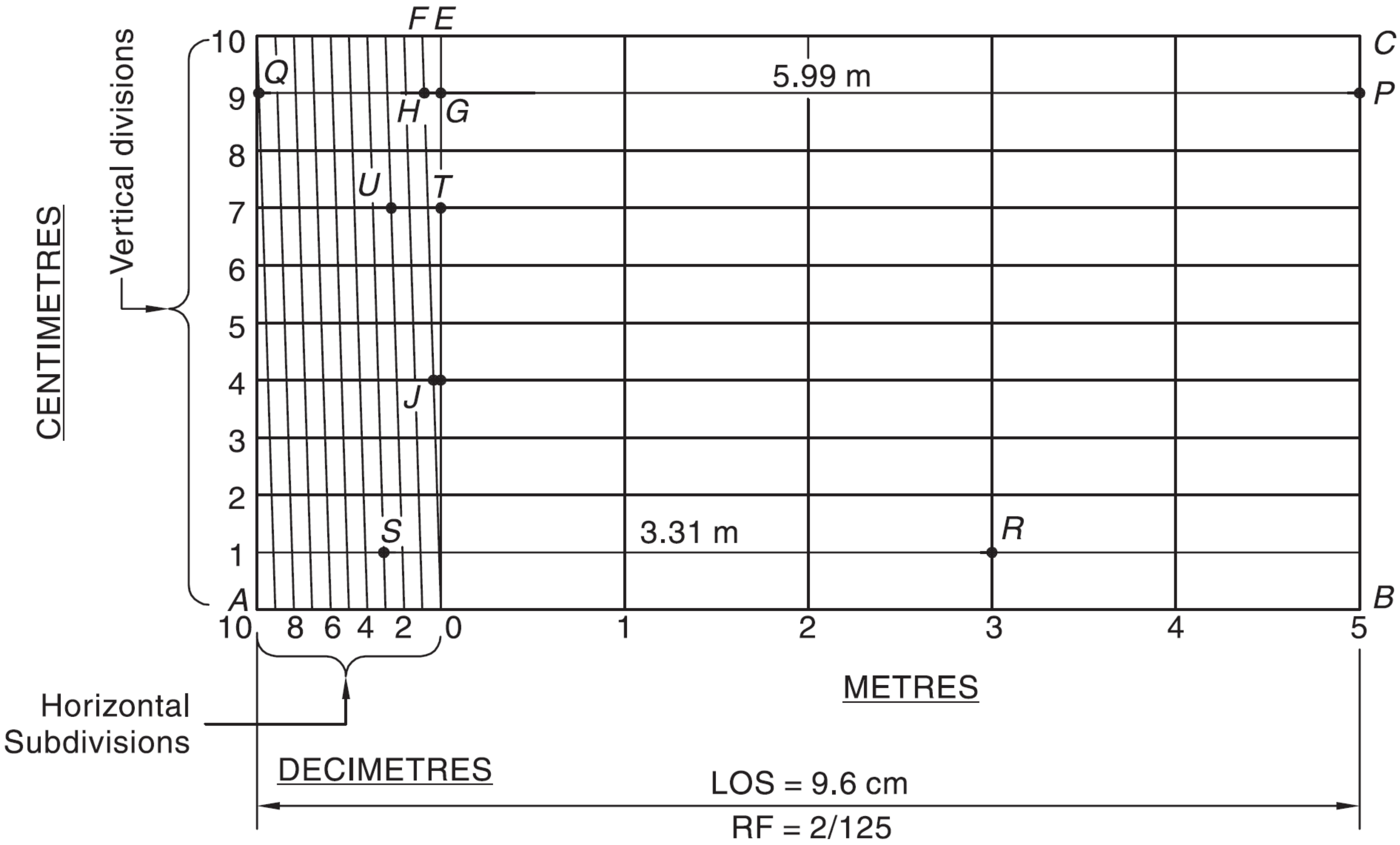
= 8 cm.

1. Draw a line 8 cm long and divide it in to 4 equal parts.
   (Each part will represent a length of 5 cm.)
2. Divide the first part into 5 equal divisions.
   (Each will show 1 cm.)
3. At the left hand end of the line, draw a vertical line and on it step-off 10 equal divisions of any length.
4. Complete the scale as explained in previous problems.
   Show the distance 13.4 cm on it.



R.F. = 1 / 2.5

DIAGONAL SCALE SHOWING CENTIMETERS.

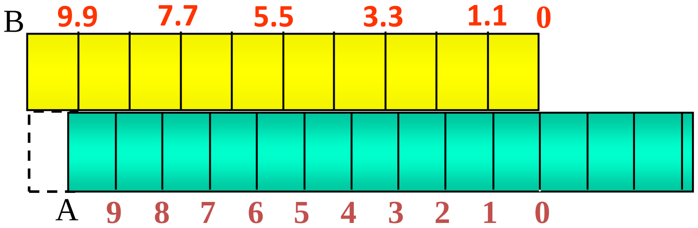


F E
10
Q
9
H
G
5.99 m
C
P
8
U
T
7
6
5
4
J
3
2
S
3.31 m
R
1
A
B
10
8
6
4
2
0
1
2
3
4
5
Vertical divisions
CENTIMETRES
Horizontal Subdivisions
METRES
DECIMETRES
LOS = 9.6 cm
RF = 2/125

Vernier Scales

The lower scale *A0* is the plain scale and the upper scale *B0* is the vernier. The combination of plain scale and the vernier is the vernier scale.

- A plain scale in which length A-0 represents 10 cm.
- Divide A-0 into ten equal parts, each will be of 1 cm.
- It would not be easy to divide each of these parts into ten equal divisions to get measurements in millimeters



Backward vernier scale

- Take a length BO equal to 10 + 1 = 11 such equal parts, thus representing 11 cm, and divide it into ten equal divisions, each of these divisions will represent 11 / 10 = 1.1 cm.
- The difference between one part of AO and one part of BO will be equal 1.1 – 1.0 = 0.1 cm or 1 mm.

Forward vernier scale

- Take a length BO equal to 10 - 1 = 9 such equal parts, thus representing 9 cm, and divide it into ten equal divisions, each of these divisions will represent 9 / 10 = 0.9 cm.
- The difference between one part of AO and one part of BO will be equal 1.0 – 0.9 = 0.1 cm or 1 mm.

- Least count (LC): Minimum length that can be measured by vernier scale.
  - LC = MSD – VSD (MSD > VSD) → Forward vernier scale
  - LC = VSD – MSD (VSD > MSD) → Backward vernier scale

Problem No. 1: The actual length of 300 m of an auditorium is represented by a line of 10 cm on a drawing. Draw a vernier to read upto 600 m and mark a length of 364 m on it.

RF = 10 cm / (300 x 100) cm = 1/3000

LOS = 600 x 100 x RF = 20 cm

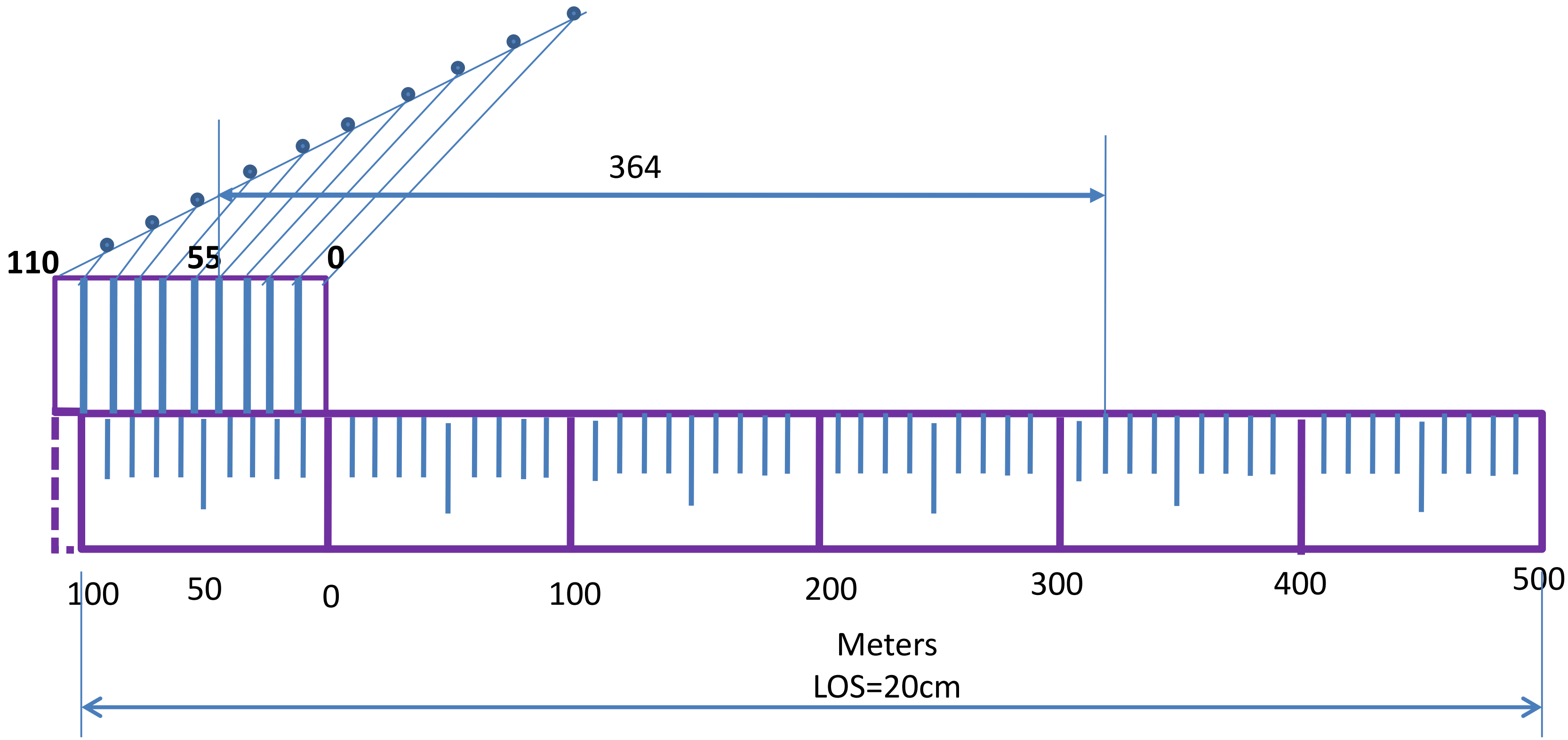
Divide scale in 6 equal parts. Each division will represent 100 m (by Construction method).

Divide each division in 10 subdivisions such that 1 sub-division = 10 m, ie, MSD = 10.

Backward/Retrograde Vernier: VSD > MSD and LC = VSD - MSD

LC = VSD-MSD => 10 VSD = 11 MSD => each sub-division on VSD = 11/10 = 1.1

Using backward vernier scale, length = 44 (VSD) + 320 (MSD)



Problem no. 2 A map of size 500 cm x 50 cm wide represents an area of 6250 sq.Kms. Construct a vernier scale to measure kilometers, hectometers and decameters and long enough to measure up to 7 km. Indicate on it a) 5.33 km b) 59 decameters.

Vernier Scale

SOLUTION:

$$RF = \sqrt{\frac{\text{AREA OF DRAWING}}{\text{ACTUAL AREA}}}$$

$$= \sqrt{\frac{500 \text{ X } 50 \text{ cm sq.}}{6250 \text{ km sq.}}}$$

$$= 2 / 10^5$$

Length of scale = RF X max. Distance

$= 2 / 10^5 \text{ X } 7 \text{ kms}$

$= 14 \text{ cm}$

CONSTRUCTION: *(Main scale)*

Draw a line 14 cm long.
Divide it in 7 equal parts.
(each will represent km)
Sub-divide each part in 10 equal parts.
(each will represent hectometer)
Name those properly.

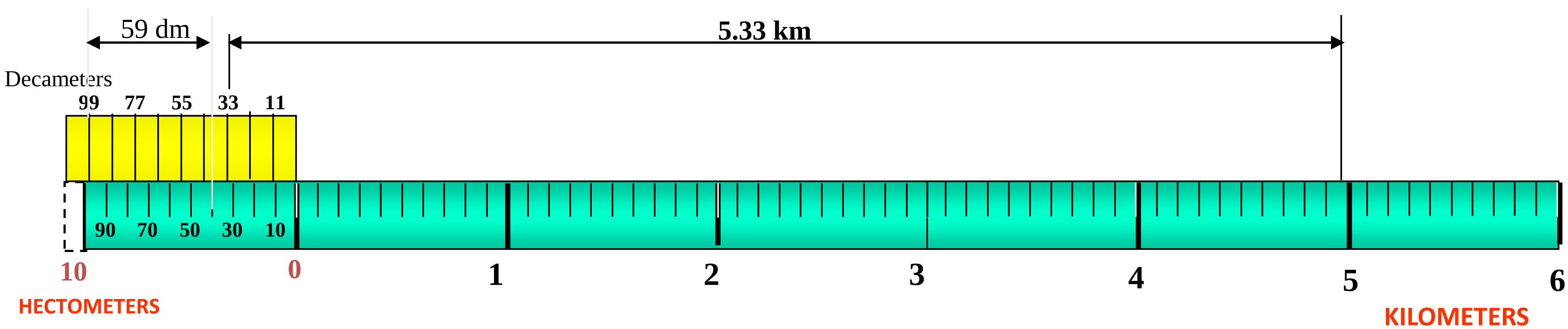
CONSTRUCTION: *(vernier)*

Take 11 parts of hectometer part length and divide it in 10 equal parts.
Each will show 1.1 hm or 11 dm and covering in a rectangle complete scale.

TO MEASURE GIVEN LENGTHS:

a) For 5.33 km :
Subtract 0.33 from 5.33
i.e. 5.33 - 0.33 = 5.00
The distance between 33 dm (left of Zero) and 5.00 (right of Zero) is 5.33 k m

(b) For 59 dm :
Subtract 0.99 from 0.59
i.e. 0.59 – 0.99 = - 0.4 km
(- ve sign means left of Zero)
The distance between 99 dm and - 0.4 km is 59 dm
(both left side of zero)



Poblem no. 3
Draw a vernier scale of RF = 1 / 25 to read centimeters upto
4 meters and on it, show lengths 2.39 m and 0.91 m

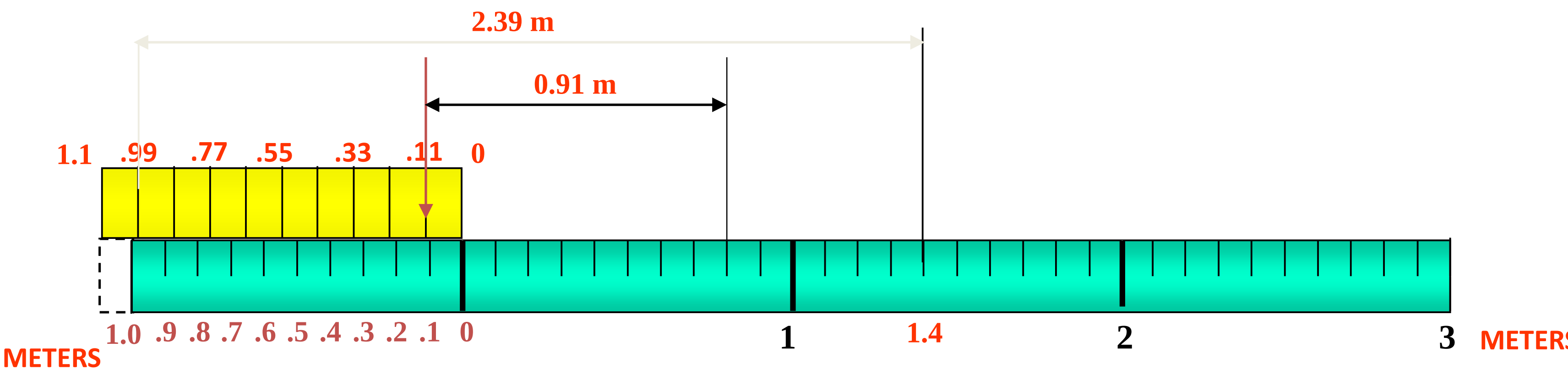
Vernier Scale

SOLUTION:
Length of scale = RF X max. Distance
= 1 / 25 X 4 X 100
= 16 cm

CONSTRUCTION: *(Main scale)*
Draw a line 16 cm long.
Divide it in 4 equal parts.
(each will represent meter)
Sub-divide each part in 10 equal parts.
(each will represent decimeter)
Name those properly.

CONSTRUCTION: *(vernier)*
Take 11 parts of Dm length and divide it in 10 equal parts.
Each will show 0.11 m or 1.1 dm or 11 cm and construct a rectangle
Covering these parts of vernier.

TO MEASURE GIVEN LENGTHS:
(1) For 2.39 m : Subtract 0.99 from 2.39 i.e. 2.39 - 0.99 = 1.4 m
The distance between 0.99 (left of Zero) and 1.4 (right of Zero) is 2.39 m
(2) For 0.91 m : Subtract 0.11 from 0.91 i.e. 0.91 – 0.11 = 0.80 m
The distance between 0.11 and 0.80 (both left side of zero) is 0.91 m



Problem No. 4: The actual length of 300 m of an auditorium is represented by a line of 10 cm on a drawing. Draw a vernier to read upto 600 m and mark a length of 364 m on it.

RF = 10 cm / (300 x 100) cm = 1/3000

LOS = 600 x 100 x RF = 20 cm

Divide scale in 6 equal parts. Each division will represent 100 m (by Construction method).

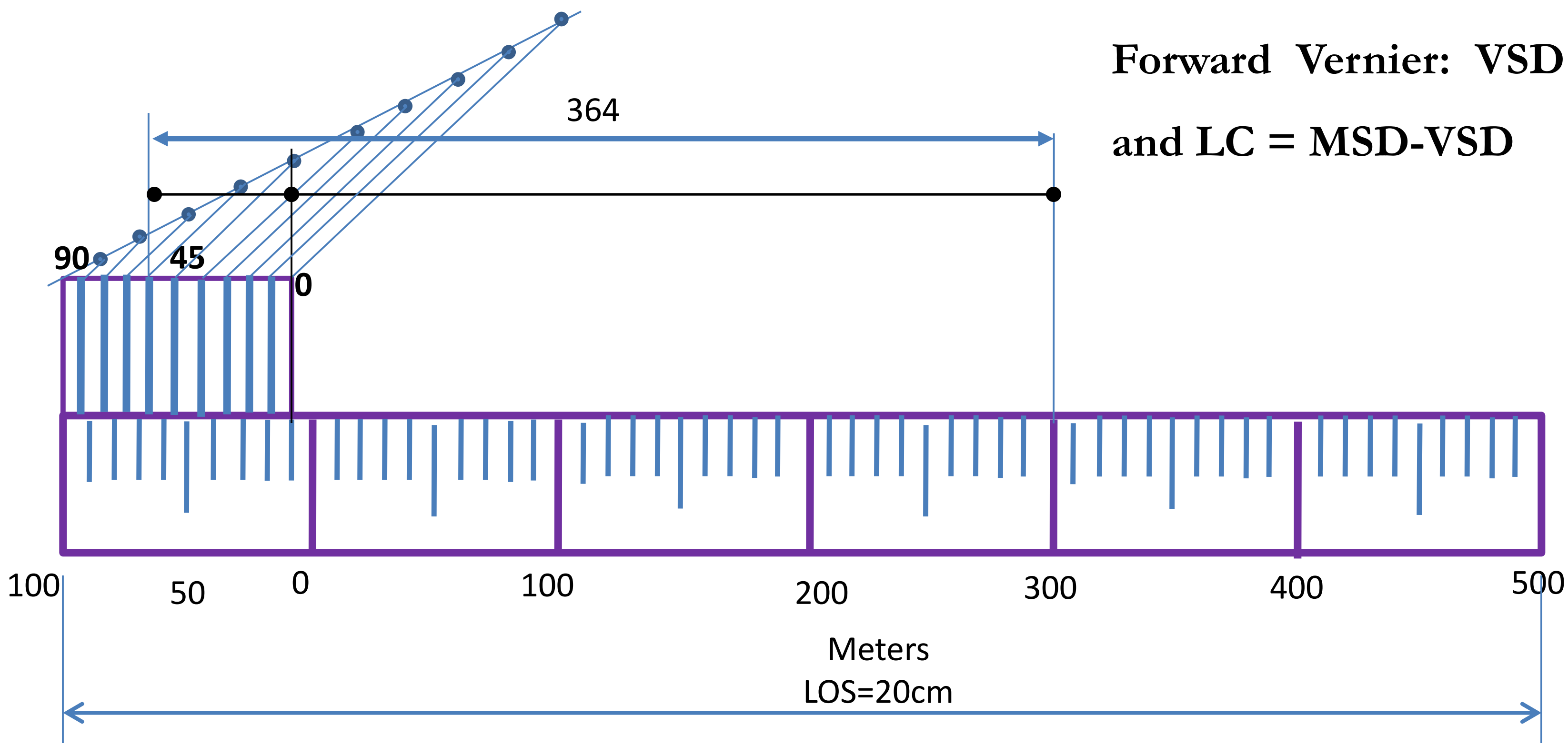
Divide each division in 10 subdivisions such that 1 sub-division = 10 m, ie, MSD = 10.

Least count (LC) = 1 m = 1/10 MSD

LC = MSD-VSD

=> 10 VSD = 9 MSD => each sub-division on VSD = 9/10 = 0.9

Using **Forward vernier scale**, length = 54 (VSD) + 310 (MSD)



Forward Vernier: VSD <MSD

and LC = MSD-VSD

Construct a vernier scale than can read centimetre and measure upto 4 meters. Its RF is 1/25 and show the length of 2.28m. Use Backward and Forward methods.

Soln:

1) RF = 1/25

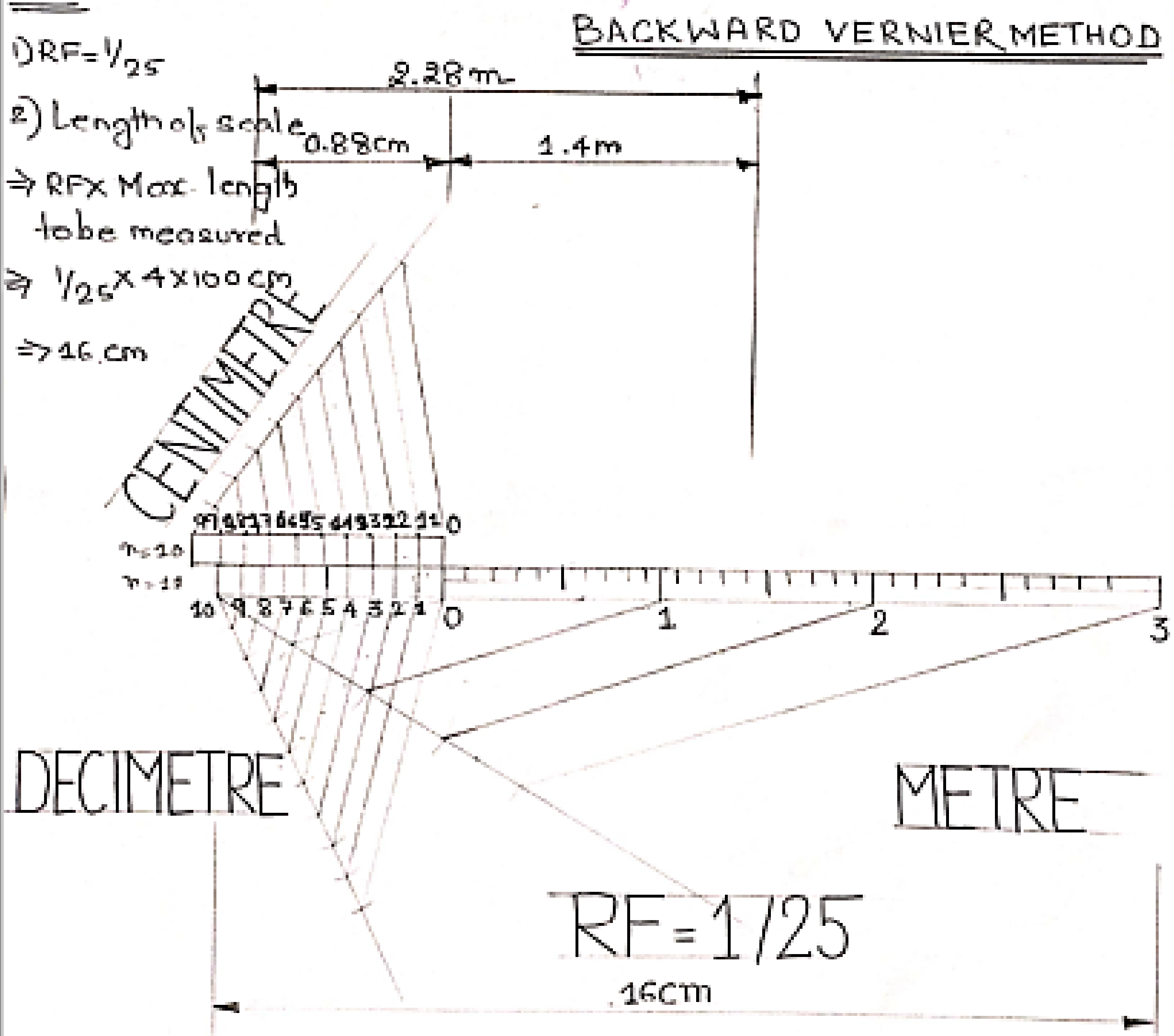
2) Length of scale

⇒ RF × Max. length to be measured

⇒ 1/25 × 4 × 100 cm

⇒ 16 cm

BACKWARD VERNIER METHOD



2.28m = 2 + 0.2m + 0.08m
= 2 + 2dm + 8cm
1m = 10dm, 1m = 100cm

For Backward Scale take

Number of main scale division = 11 and Number of vernier scale division = 10

Number of main scale division
= Number of Vernier scale division for same length

MSD size < VSD size

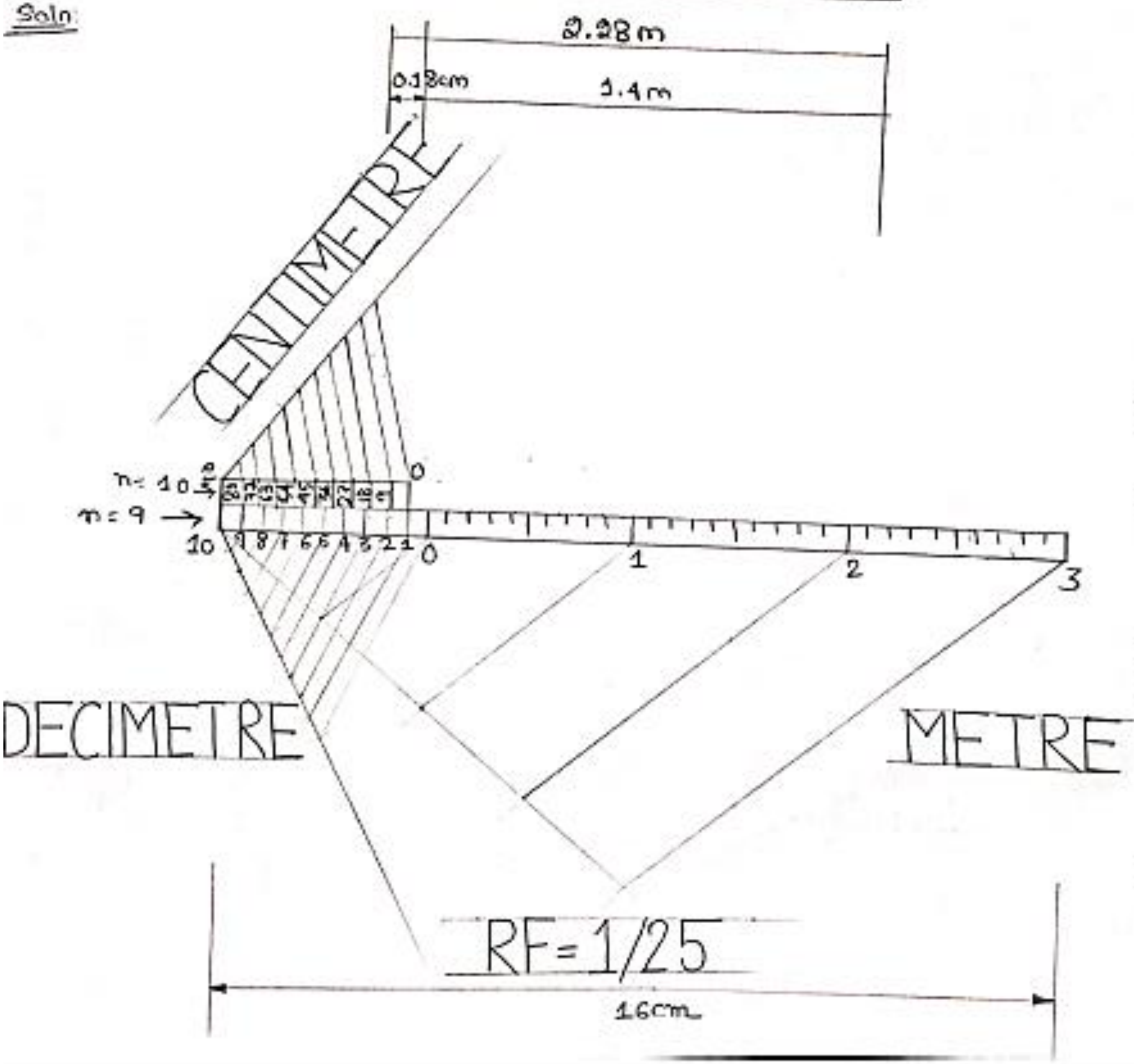
$0.1m < \frac{1.1}{10} = 1.1dm = 0.11m = 11cm$

2.28 = 0.88(VS) + 1.4(ms)

Least count for Vernier scale
= VSD − MSD = 0.11m − 0.1m
= 0.01m = 1cm

FORWARD VERNIER METHOD

Soln:



MSD Size > VSD Size

MSD value > VSD value

0.1m $\quad \frac{9}{10}$ dm

= 0.9 dm

= 0.09 m (0.9/10)

= 9 cm

For forward vernier scale take

No. of division in main scale = 09

No. of division in vernier Scale = 10

No. of division of main scale = No. of vernier scale div. for the same length

2.28 = 0.18(VS) + 2.1(MS)

Least count for vernier scale

= MSD − VSD = 0.1 − 0.09

= 0.01 m

= 1 cm

Thank You for Patient Hearing